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Cybersecurity in Satellite Communication Networks: Key Threats and Neutralization Measures

ADITI RANJIT KUMAR VERMA (Senior Member, IEEE)

Department of Messaging & Application Quality Engineering, T-Mobile USA Inc., Bellevue, WA 98005, USA

CORRESPONDING AUTHOR: A. R. K. VERMA (e-mail: aditi.r.verma@gmail.com)

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ABSTRACT The proliferation of satellite mega-constellations, such as SpaceX's Starlink and NewSpace initiatives, particularly CubeSat deployments, and their integration into terrestrial 5G/6G networks have transformed space-based communications. While this technological convergence has brought numerous advantages, it has also introduced a new vector of cybersecurity threats to both traditional satellite infrastructure and novel small-satellite architectures. Owing to their broadcast nature and extensive coverage area, satellite signals are inherently susceptible to jamming, spoofing, and eavesdropping, rendering them appealing targets for malicious actors. Moreover, satellites operate under power, computational, and bandwidth limits, which restrict the implementation of resource-intensive security measures. Therefore, it is necessary to develop robust and lightweight security frameworks capable of countering threats across multiple orbital domains, communication protocols, and network integration points. This paper presents a hybrid research contribution that combines a systematic survey of key cyber threats with novel conceptual and practical security frameworks for satellite communication (SATCOMs) networks. It weaves threats, defenses, and governance into a coherent scaffold that follows the entire SATCOM stack–space, ground, and user–across all orbits. Unlike most previous studies–which treat countermeasures in a fragmented manner with limited synergy across layers–this study presents cyber threats using the confidentiality-integrity-availability (CIA) triad and presents a multilayered security framework that combines physical-layer hardening, quantum-resilient encryption, and AI-driven analytics, supported by a governance framework aligned with standards, within one integrated workflow. The undertaken approach uniquely addresses prevention, defense, resilience, and collaboration through detailed countermeasures. This bridges the gap between isolated security solutions and the interconnected nature of modern SATCOM ecosystems. The analysis of the research reveals that prioritizing lightweight AI, quantum resilience, and cross-layer security by real-world SATCOM operators and agencies is critical for securing next-generation SATCOMs against sophisticated cyber-attacks. Moreover, the convergence of SATCOMs with 5G/6G, AI, and quantum technologies requires interdisciplinary research to address the interoperability, scalability, and ever-increasing cyber threats in SATCOMs.

INDEX TERMS Satellite communications (SATCOM), cybersecurity, cryptographic methods, satellite jamming, spoofing.

I. INTRODUCTION

SATCOM networks are the cornerstone of modern telecommunications infrastructure, facilitating an extensive array of mission-critical services across the globe. As of early 2024, approximately 9500 satellites have been launched into orbit [1], including communication, Earth observation, navigation, military, and space debris satellites. Traditional applications of SATCOMs include global positioning system (GPS) navigation [2], broadband connectivity in geographically isolated regions [3], [4], radio frequency (RF) broadcasting, meteorological forecasting [5], military communications [6], and maritime operations [7]. However, modern business endeavors by tech giants such as Facebook, SpaceX, and Amazon [8] have garnered renewed interest in SATCOMs with a range of new applications (e.g., to enable ultra-reliable and low-latency communications

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(URLLC) for time-critical applications [4] and low-power Internet of things (IoT) connectivity [9]. Even the 'big data'[1] phenomenon [10] heavily involves satellites; the EU/ESA Earth observation program is among the world's largest big data providers[11]. It has been estimated that the SATCOM market share will expand from USD 31.22 billion in 2023 to USD 70.89 billion in 2032, with a compound annual growth rate (CAGR) of 9.6 percent [12], [13]. A typical SATCOM network comprises different interconnected components that collaborate to facilitate uninterrupted communication between ground-based stations, orbiting satellites, and end-user devices. SATCOMs are further considered the backbone for enabling 5th generation (5G) and forthcoming 6th generation (6G) technologies, thereby contributing to the next generation of services, communications, and networks [11]. The integration of hybrid satellite-terrestrial networks, formally recognized through the European Space Agency (ESA) and industry collaboration on 'Satellite for 5G' [11], expands the reach and reliability along with global connectivity. This integration was further strengthened in 2019 when the ESA established the 'Space for 5G and 6G' programs [11]. The 6G architecture is expected to incorporate both terrestrial and non-terrestrial networks (NTNs). This includes satellites, in order to achieve data rates from 100 Gb/s to 1 Tbps as well as minimize latency to 0.1 ms Integration will also improve other metrics, such as energy efficiency, positioning accuracy, and connectivity density [4]. SATCOMs are essential for space-air-ground integrated networks (SAGIN), terahertz communication, and quantum communication (QC) [4], [14], [15]. Fig. 1 depicts various SATCOM applications across different strata of life. It shows the range of SATCOM applications across urban, rural, and remote areas and highlights services such as media broadcasting, mobile IP connectivity, and emergency response. Urban areas rely on SATCOM for mobile networks, Internet Protocol (IP) backhauls, and media distribution. Rural and remote regions use it for maritime tracking, field communication and disaster relief. It includes sectors such as government, aviation, energy, and IoT.

SATCOMs show great promise for diverse communication channels and applications [16]. However, incorporating them into contemporary cyberspace has increased the system's susceptibility to space- and cyber-threat attacks for both non-governmental entities, such as civilians, and governments [11]. This potential compromise may lead to the breakdown of multiple essential services [17]. Cyber-attacks are thought to be the crime with the fastest rate of growth globally. They are becoming larger, more complex, and more expensive [11], and space assets remain at risk from these rapidly developing technologies [18]. Several threat actors, such as nation-state militaries, cyber adversaries, and commercial rivals, have made SATCOM networks their preferred targets [19], [20].

Vulnerabilities in SATCOM networks, such as their distributed nature [13], inadequate authentication procedures, antiquated software, inadequate encryption techniques, and improper network isolation, are exploited by these entities. Satellite signals are susceptible to jamming or interception owing to the intrinsic nature of wireless communications, which raises further security issues [21]. Typical threat ranges include, but are not limited to, eavesdropping, tampering, disruption, and jamming [13], [22]; intrusions from traditional Internet-based threats such as malware and denial of service (DoS) attacks [23]; and patching vulnerabilities [24]. The potential dangers posed by these threats could be even more significant, considering that many current SATCOMs either lack security measures entirely or employ outdated security protocols [13]. These systems are ill-equipped to handle sophisticated attacks prevalent today [25]. Moreover, there is a lack of international government oversight of cybersecurity protocols for space-based assets [19]. Although a few nations have implemented domestic policies governing the oversight of SATCOMs, effective global regulation with supervisory capabilities remains elusive, with the United Nations (UN) and the United Nations Office of Outer Space Affairs (UNOOSA) possessing very limited regulatory influence [26]. This lack of global regulation and universal cybersecurity standards makes SATCOMs networks susceptible to cyberattacks. There have been multiple instances of hostile cyber-attacks that have disrupted SATCOMs and violated international law [11].

Considering that SATCOM networks and cyberspace operations are worldwide in scope, it is vital to protect these space assets. Recognizing this need, significant contributions have been made in the past decade. To curb eavesdropping, jamming, spoofing, and DoS threats, engineers have reinforced the physical layer with specialized security measures, cryptographic techniques, anti-jamming methods, blockchain workflows, and hardened hardware solutions. Against attacks that target space, ground, and link segments, such as satellite hijacking, unauthorized access, and data manipulation. However, a few weaknesses remain in the deployment of traditional cryptographic methods. Most SATCOM links, for instance, use encryption weaker than 128 bits[2] [27] —or, in some cases, none at all–and often rely on fixed keys that, once compromised, allow adversaries to read past and future transmissions [28]. This is due to the fact that onboard processors are limited in power and are susceptible to radiation, which hinders their ability to manage complex mathematical codes. Furthermore, the imminent advent of practical quantum computing could render Rivest–Shamir–Adleman (RSA) and

[1]Big data is characterized as vast amounts of both organized and unstructured data that are too large to handle with conventional database and software methods.

[2]Unraveling cybersecurity Challenges in Satellite Communication Systems.

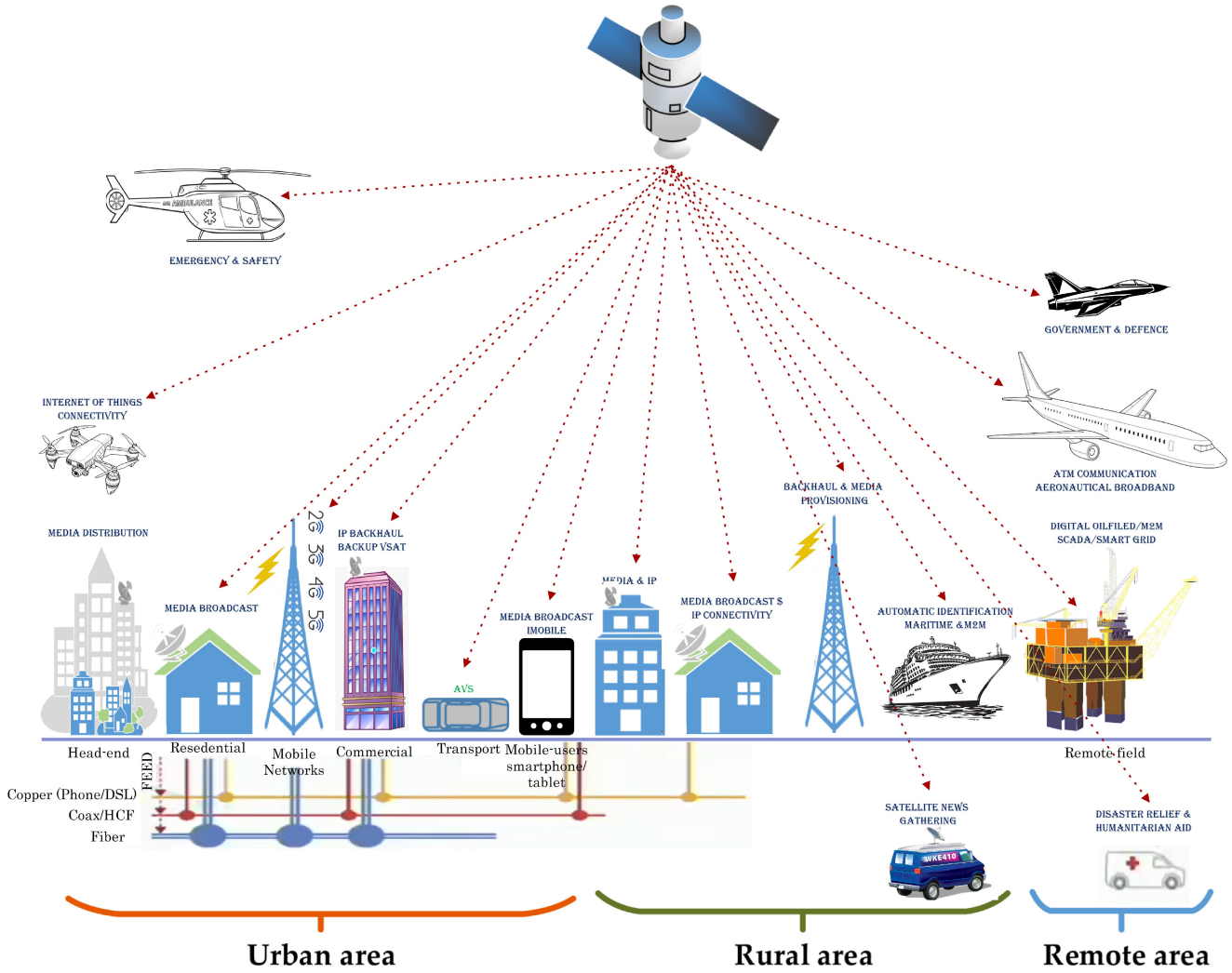


FIGURE 1. SATCOM applications in the new space era.

Elliptic Curve Cryptography (ECC) algorithms ineffective during the operational lifespan of many spacecraft [22]. DNA cryptography has been proposed to counter the inherent weaknesses of traditional cryptographic methods [29]. It leverages the complexity of biological systems, adding a security layer that takes advantage of the biological difficulties in DNA manipulation [30]. The extraordinary storage density of DNA–approximately 1 g can store 455 exabytes of data–makes it ideal for resource-constrained satellite environments [31]. The flexibility in key generation, where senders can choose any combination of DNA bases during the encryption process, significantly improves data security through an increased key space [32]. Crucially, the difficulty of manipulating or brute-forcing nucleic acid sequences in DNA cryptography confers resistance to quantum algorithms, such as those proposed by Shor and Grover [33]. When the computationally intensive stages of sequencing, error correction, and one-time key generation are delegated to elastic cloud resources, operators can administer scalable, policy-driven key orchestration across large constellations without burdening radiation-hardened avionics [32]. Consequently, a cloud-enabled DNA-cryptographic framework provides a durable, low-overhead defense that surpasses the protective capacity of conventional purely mathematical schemes in safeguarding SATCOMs. Section I-A briefly captures our unique research contribution and structure of the paper, and Section I-B provides a brief overview of existing works.

A. CONTRIBUTION AND ROAD MAP

Previous studies have established fundamental approaches in areas such as PLS, cryptographic protocols, and network architectures for SATCOMs, as discussed in Section I-B. However, the current SATCOM cybersecurity literature is fragmented; it lacks a unified taxonomy spanning LEO-MEO-GEO and the space-ground-user triad [17], [22], offers only piecemeal countermeasures with little cross-layer or post-quantum/AI integration [11], [34], confines policy discussion to high-level recommendations without actionable governance paths [11], [13], and provides no

forward-looking roadmap for AI-enhanced threat intelligence or post-quantum assurance [13], [22]. This study presents a more comprehensive end-to-end security framework for SATCOMs that extends beyond orbital assets to include ground stations, links, and user segments. It offers a holistic threat analysis and multi-domain neutralization strategy–spanning prevention, defense, resilience, and collaboration–thereby providing a novel, operationally viable paradigm beyond theoretical analysis that significantly advances the state of SATCOMs security. In summary, the following dual contributions are offered to the field. First, a comprehensive survey of existing SATCOM cyber vulnerabilities across space, ground, and user segments is conducted, and second, practical and conceptual proposals for integrated security frameworks that address the identified vulnerabilities are put forward. This study integrates space, ground, and user segments into one system and explores various practical ways in which AI-driven detection and post-quantum encryption can support active defense. It presents practical solutions for real-world SATCOM operators and agencies to protect their systems from cyber threats. It reviews the risks, countermeasures, and defense tactics used during system development and deployment. It also addresses the challenges posed by satellite mega-constellations and examines how these networks intersect with 5G and 6G technology. Compared to more recent studies, such as Radanliev [35] which focus on integrating CHERI-based hardware security with AI Bills of materials to enhance software supply chain transparency and memory safety in AI systems, our work takes a broader systems-level view of SATCOM security. Importantly, this study shifted from conceptual framing to deployment-ready security architectures, particularly by embedding AI-driven detection and post-quantum cryptography in operational SATCOM environments. The following are the major contributions of this study:

1) Develop a tri-layer threat taxonomy for low earth orbit (LEO), medium earth orbit (MEO), geostationary equatorial orbit (GEO) constellations that links physical (ASAT, directed-energy), communications (jamming, spoofing), and service-availability (DoS/DDoS) attacks into one unified attack surface spanning space, ground, and user segments.
2) Analyze a five-domain defense stack that couples (i) adaptive beamforming to harden physical links, (ii) a machine learning and deep-learning intrusion detection (iii) a hybrid QKD–PQC scheme for post-quantum cryptographic agility, (iv) scalable multi-antenna hardware front-ends, and (v) integrated AOA–Doppler–NMA–SSSC processing to suppress spoofing. Together, these elements constitute an end-to-end protection blueprint for next-generation satellite networks.
3) Evaluate a security-governance blueprint that harmonizes WRC-23, ISO/TS 20517:2024, and the NATO 2022 Strategic Concept with concrete standardization pathways through CCSDS, 3GPP-NTN, and GSE. The framework couples novel public–private partnership models for infrastructure resilience with a layered DoS/DDoS response playbook and maps future research priorities in post-quantum assurance and AI-enhanced threat intelligence.

The remainder of this paper is arranged as follows:

Section II introduces the basics of SATCOM networks, such as the communication framework and essential satellite characteristics in the LEO, MEO, and GEO. Section III discusses the cybersecurity imperatives for modern SATCOM networks in the confidentiality, integrity, and availability (CIA) triad. Section IV discusses the most popular threat vectors to modern SATCOM networks and outlines various novel and established cyber threat neutralization approaches. Finally, in Sections V and VI we analyze some of the prospects for future research and provide concluding remarks, respectively.

B. TRENDS AND EVOLUTION OF DISCUSSION IN SATCOMS AND CYBER-SECURITY

To maintain the operational integrity of SATCOMs, securing SATCOM links across space, ground, and user segments is crucial. Over the years, research has focused on resilient communication frameworks that can withstand cyber threats and provide secure communication channels. From the initial focus on legal and regulatory frameworks to addressing specific cyber-attacks and the development of countermeasures, the need to protect the increasingly interconnected and vital SATCOM network has become more pronounced. Currently, the use of innovative technologies to address emerging challenges is being explored. For instance, quantum cryptography, which is capable of near-completely unbreakable encryption, has emerged as a promising approach for securing SATCOM networks [17], [36]. Similarly, the integration of AI has become imperative, thereby enhancing threat detection and response systems for more adaptive and robust cybersecurity measures [36]. These advancements reflect the evolving nature of SATCOM cybersecurity, driven by technological innovation and the need to protect increasingly interconnected and vital infrastructure. In particular, contemporary advancements have addressed various aspects of security. They seek physical layer security (PLS) [37], [38], [39], enhance cryptographic protection measures [40], [41], [42], and amalgamate terrestrial and space network architectures [34], [43], [44].

Tables 1 and 2 provide a brief overview of the selected studies conducted in the domain of SATCOM networks. Threats such as jamming, spoofing, eavesdropping, and DoS are outlined, and countermeasures, including cryptographic methods, intrusion detection, AI, and SDN, are discussed. It should be noted that most studies rely on literature reviews or qualitative analyses, and only a few apply or test solutions in operational environments. Satellite types include LEO, MEO, GEO, and CubeSats, and the application areas span military,

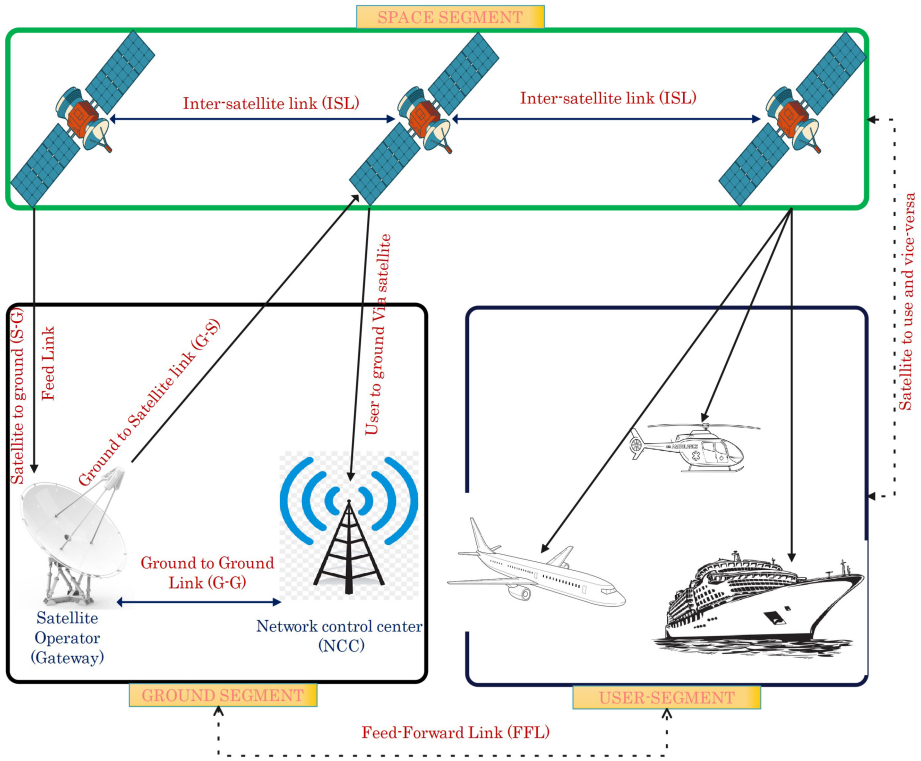


FIGURE 2. Overview of SATCOMs links across space, ground, and user segments.

commercial, and critical infrastructure domains. Table 3 lists the acronyms.

II. BACKGROUND

The network architecture of SATCOMs comprises distinct but interconnected parts of space-based and terrestrial technologies that operate across multiple frequency bands and orbital configurations to deliver reliable communication services worldwide. This section presents the fundamental concepts associated with the SATCOM networks utilized in our study, including satellite constellations, structures, and relevant protocols. The primary objective was to equip the audience with essential background and supplementary knowledge of SATCOM technologies and their key characteristics.

A. COMMUNICATION FRAMEWORK

A typical SATCOM network reference communication architecture is illustrated in Fig. 2, consists of three main segments - space, ground and user segments - which interact with one another via radio frequency (RF) signals [45]. The space segment consists of artificial satellites in orbit, which are categorized by their functions and orbital positions [46]. Each orbit serves a distinct purpose and offers unique benefits in terms of coverage, latency and capacity. These satellites primarily function as orbital repeater stations, receiving signals from ground stations (GSs) and retransmitting them to designated locations on Earth [47]. An artificial satellite typically consists of two elements: a payload and a bus. The payload is responsible for the specific tasks/functions that the satellite is designed to perform and houses the communication equipment [48]. SATCOM networks utilize RF waves to communicate with other satellites, typically operating within the MHz and GHz frequency ranges [49]. The communication payload architecture integrates multiple transponders operating across the C, Ku, and Ka bands with frequencies ranging from (4-8 GHz), (12-18 GHz) and (26.5-40 GHz), respectively [13]. Moreover, signals are sent from Earth to the satellite via an uplink channel, whereas a downlink channel facilitates signal transmission from the satellite back to Earth.

To reduce interference both terrestrially and in orbit, SATCOM networks typically employ distinct frequencies for uplink and downlink channels [50]. For communications in general, including SATCOMs, the bands, regulation of frequencies, and recommendations have been authorized by both the Federal Communications Commission (FCC) and the International Telecommunication Union (ITU) [51]. The other component of the satellite, the bus, supports the payload with power, propulsion, and thermal control systems [25]. It is responsible for executing all essential functions that enable the satellite to maintain its orbital position and support its payload [48], [52]. This system encompasses various subsystems, including power generation and distribution [53], thermal regulation [54], communication networks [46], attitude and orbit control mechanisms, onboard computers, and other components.

TABLE 1. Literature Review Summary (Part I).

Author(s) & Year	Key Threats	Countermeasures	Research Method	Network Layer Focus	Types of Satellites	Application Domains	Research Gaps
Minjae Kang, Sungbin Park, Yeonjoon Lee et al. (2024) [22]	Eavesdropping; jamming; spoofing; DoS/DDoS; PL attacks	PL security; cryptographic methods; anti-jamming techniques; blockchain; hardware security	Literature review	PL; Data Link Layer; Network Layer	LEO; MEO; GEO; CubeSats	Military; commercial; scientific	In-depth technical details are lacking, with no comparative analyses of specific neutralization techniques; proposed countermeasures are discussed without extensively addressing the practical challenges of their implementation.
Sara Salim, Nour Moustafa, Martin Reisslein et al. (2024) [17]	Cyber-attacks on space segment (hardware/software attacks, satellite hijacking, signal jamming); cyber-attacks on ground segment (unauthorized access, data manipulation, DoS; cyber-attacks on links segment	Authentication and encryption methods; software updates and patch management; intrusion detection and anomaly detection systems	Comprehensive literature review	Network Layer vulnerabilities in SATCOMs; Ground and links segments	GEO; LEO satellites	Military; maritime; urban air mobility	Insufficient exploration of how policies directly address technical vulnerabilities in SCSs; Sparse analysis of attack likelihood/impact across SCS types (e.g., LEO vs. GEO); inadequate coverage of dynamic, evolving cyber threats targeting SCSs.
Dr. Antonio Carlo and Dr. Kim Obergfaell (2024) [11]	Cyber-attacks on critical infrastructures and SATCOMs; increasing sophistication and targeting of industrial control systems	Risk-based approach to enhance protection and resilience; comprehensive security measures, legal and policy frameworks; security-by-design approach in engineering; development of flexible and secure reference architectures	Qualitative analysis	Hybrid satellite-terrestrial networks; integration of space with 5G technology to enhance global connectivity and resilience	GEO; complex SATCOMs	Commercial (telecommunications); scientific research; military; critical infrastructure (energy, transportation, administration)	This study fails to provide concrete policy recommendations for international cooperation on space cybersecurity standards and enforcement mechanisms. In addition, the role of AI and modern methods in combating cyber threats is lacking, and sufficient weight is given to the legal approach to counter cyber threats.

The ground segment encompasses the terrestrial infrastructure necessary to control satellites and manage communications services. The ground segment can facilitate the communication link between satellites and user segment terminals. This component encompasses specialized gateway stations, including satellite operators, control infrastructure,

TABLE 2. Literature Review Summary (Part II).

Author(s) & Year	Key Threats	Countermeasures	Research Method	Network Layer Focus	Types of Satellites	Application Domains	Research Gaps
Pingyue Yue et al. (2023) [34]	Eavesdropping, jamming, spoofing, message modification, DoS, node compromise, and node destruction	Examined both active and passive security solutions categorized by prevention, detection, and mitigation approaches while analyzing key trade-offs and extracting practical lessons from these reliability enhancement mechanisms.	Literature review	PL., connection, and service layers for integrated space-terrestrial systems	LEO	Military: surveillance, secure communication, and navigation systems	Focusing only on satellites in orbit narrows this perspective. Ignoring ground stations, links, and user devices leaves major gaps in how we understand and secure the full space communications ecosystem.
Pietro Tedeschi, Savio Sciancalepore, Roberto Di Pietro et al. (2022) [13]	Eavesdropping; GNSS spoofing; GNSS jamming	Information-theoretic security approaches; anti-jamming techniques; anti-spoofing measures; cryptographic protocols (e.g., AES)	Comprehensive survey and literature review	Link-layer security in SATCOMs	LEO; MEO; GEO	SATCOMs; GNSS applications; emerging technologies in 6G initiatives	The manuscript remains predominantly conceptual; it offers neither experimental validation nor case-based analysis and provides no benchmarking framework for comparing the proposed countermeasures. Although artificial intelligence, quantum-resilient security, and intelligent reflecting surfaces have been cited, their operational mechanics and integration pathways have not been elucidated.
Oltjon Kodheli et al. (2021) [46]	Various challenges in SATCOMs, including security vulnerabilities, interference, and the integration of satellite systems with terrestrial networks	Development of advanced encryption techniques; robust network protocols; implementation of software-defined networking (SDN) to enhance security and efficiency	Comprehensive literature review	Medium access control techniques; networking protocols; integration of SATCOMs within the 5G ecosystem	GEO; MEO; LEO satellites	5G integration; earth observation; space communications; aeronautical and maritime tracking and communication	The paper lists threats and countermeasures but skips real-world validation; it does not test any solutions through simulation or field data; no framework combines AI, blockchain, or SDN for active defense; and does not offer practical steps for applying policies in space cyber defense.

and network operator facilities, such as network management centers (NMC) and network control centers (NCC). These elements support and manage satellite access requests from the users.

The user segment encompasses various user terminals, including satellite mobile phones, vessels, and aircraft, from fixed very small aperture terminals (VSATs) to mobile aeronautical systems. These devices connect to satellites through

TABLE 3. Acronym list.

Abbreviation	Full Form
3GPP	3rd Generation Partnership Project
ABSC	Ambient Back-Scatter Communication
ADS	Active Defense Systems
AOA	Angle of Arrival
AN	Artificial Noise
ASAT	Anti-Satellite
BiLSTM	Bidirectional Long Short-Term Memory
CAGR	Compound Annual Growth Rate
CDMA	Code Division Multiple Access
CNN	Convolutional Neural Networks
CSpO	Combined Space Operations Initiative
CR	Cognitive Radio
DDoS	Distributed Denial-of-Service
DARPA	Defense Advanced Research Projects Agency
DL	Deep Learning
DNN	Deep Neural Networks
DoS	Denial-of-Service
DRAB	Dynamic Rotated Angular Beamforming
ESA	European Space Agency
FCC	Federal Communications Commission
FHSS	Frequency Hopping Spread Spectrum
FL	Federated Learning
FSS	Fixed Satellite Service
GEO	Geostationary Equatorial Orbit
GNSS	Global Navigation Satellite Systems
GPS	Global Positioning System
GSs	Ground Stations
HAPS	High-Altitude Platform Systems
HELLADS	High-Energy Liquid Laser Area Defense Systems
IoT	Internet of Things
ITU	International Telecommunication Union
JSDFH	Jamming Statistics-Dependent Frequency Hopping
LEO	Low Earth Orbit
MEO	Medium Earth Orbit
ML	Machine Learning
NBJs	Narrowband Jammers
NIST	National Institute of Standards and Technology
NMA	Navigation Message Authentication
NMC	Network Management Center
NCC	Network Control Center
PLS	Physical Layer Security
PRD	Pseudo-Range Differences
PRS	Pseudo-Range Sums
QC	Quantum Communication
QBER	Quantum Bit Error Rates
QKD	Quantum Key Distribution
RF	Radio Frequency
RFMLS	Radio Frequency Machine Learning Systems
S2G	Satellite-to-Ground
SAGIN	Space-Air-Ground Integrated Networks
SATCOMs	Satellite-based Communication
SB-QKD	Satellite-Based Quantum Key Distribution
SAR	Synthetic Aperture Radar
SDN	Software-Defined Networking
SDR	Software-Defined Radios
SGD	Stochastic Gradient Descent
SNR	Signal-to-Noise Ratio
SSA	Space Situational Awareness
SSSCs	Spread Spectrum Security Codes
THAAD	Terminal High Altitude Area Defense
UAV	Unmanned Aerial Vehicle
UNOOSA	United Nations Office of Outer Space Affairs
URLLC	Ultra-Reliable and Low-Latency Communications
VSATs	Very Small Aperture Terminal
WBJs	Wide-Band Jammers

ground-to-user pathways, including forward links [55]. They can also interact with gateways using any available communication technologies. The forward link comprises both an uplink (from the base station to the satellite) and a downlink (from the satellite to the mobile user). In contrast, satellite networks such as Globalstar, Inmarsat [56], Iridium [51], [57] and Thuraya enable user handsets to connect directly with satellites through the user-to-satellite (US) link, which typically operates on L-band frequencies [13].

Although the preceding discourse addresses the conventional SATCOM framework, numerous variations exist. Notably, in recent years, the 3rd Generation Partnership Project (3GPP) has issued multiple standards aimed at delineating the overarching architecture of non-terrestrial satellite networks within the context of 5G systems. These guidelines are intended for various forms of extraterrestrial communication, encompassing global navigation satellite systems (GNSS), high-altitude platform systems (HAPS), and air-to-ground transmissions [13]. It should also be kept in mind that cyberattacks may be initiated in any of the above-identified segments.

B. SATELLITE CONSTELLATION CHARACTERISTICS

The orbit of a satellite is characterized by several key attributes: orbital geometry (elliptical or circular), distance from Earth (low-Earth, medium-Earth, or geostationary), direction of rotation (clockwise or anticlockwise), and angle relative to the Earth's equatorial plane [58]. Among these characteristics, altitude is the most commonly referenced, leading to the classification of orbits into three main categories: LEO, MEO, and GEO. Table 4 presents a comprehensive comparison matrix of the satellite constellations.

III. CYBERSECURITY IMPERATIVES IN MODERN SATCOM INFRASTRUCTURE

The confidentiality, integrity, and availability (CIA) triad represents the security framework for modern SATCOM networks. This triad provides essential protection for information and ensures the robustness of SATCOM networks. Confidentiality ensures that information is accessible only to those with proper authorization. Integrity, on the other hand, ensures the accuracy and reliability of the data. Similarly, availability guarantees that information and systems remain accessible to authorized users when required. To ensure the CIA triad, SATCOM networks must incorporate strategies such as redundancy, failover mechanisms, and anti-jamming techniques. Fig. 3 refers to the key Components of cyber-security in SATCOM networks.

A. INTERNATIONAL SATELLITE CYBERSECURITY REGULATIONS & STANDARDS (2022–2025)

Between 2022 and 2025, the international framework for satellite cybersecurity tightened substantially. The ITU's WRC-23 Resolution 676 instructs all administrations to detect and suppress GNSS jamming or spoofing that threatens the radio-navigation satellite service (RNSS).[3] ISO followed with ISO/TS 20517:2024, the first global management standard embedding cybersecurity requirements across every phase of a space mission, from spacecraft to ground segment.[4] The UN General Assembly's Resolution 78/20 created an open-ended working group to draft norms that protect critical space

[3]GPSIA Statement on Disruptions to Global Navigation Satellite Systems.
[4]SO/TS 20517:2024 Space Systems – Cybersecurity Management Requirements and Recommendations.

TABLE 4. Comparison of satellite orbital characteristics.

Characteristics	LEO	MEO	GEO
Altitude Range	500-1,200 km	2,000-20,000 km	36,000 km
Orbital Period	85-120 min (11-17 rounds/-day)	120-480 min	24 hours
Propagation Delay	∼30 ms	∼125 ms	∼600 ms
Earth Coverage	Small	Large	Very Large
Coverage Area Per Satellite	Small footprint, requires constellation	Medium footprint	∼1/3 of Earth's surface
Data Loss	Low	Low	Medium
Required Satellites	Tens to Hundreds	At least 6	At least 3
Gateway Configuration	Numerous Flexible	Local Flexible	Stationary
Signal Strength	Strongest due to proximity	Moderate	Weakest (needs more power)
Operational Challenges	Atmospheric drag, Space debris risk, Complex constellation management	Van Allen radiation exposure, Limited visibility (2-8 hrs), Complex tracking	Launch complexity, High power requirements, Poor polar coverage
Primary Applications	High-speed internet, Mobile communications, Real-time services, Earth observation	Navigation systems (GPS), Medium-latency communications, Regional coverage	Broadcasting, Fixed communications, Weather monitoring
Lifespan Factors	Shorter due to drag, Debris collision risk, 5-7 years typical	Radiation exposure, 10-15 years typical	15-20 years typical
Cost Considerations	Lower per satellite, Higher constellation cost, Frequent replacements	Moderate launch costs, Medium replacement rate	Highest per satellite, Lowest constellation cost
Communication Characteristics	Lowest latency, High bandwidth, Frequent handovers	Medium latency, Good bandwidth, Periodic handovers	Highest latency, Stable connections, No handovers
Coverage Limitations	Small footprint, Requires many satellites, Global with constellation	Good regional coverage, Moderate gaps, Multiple satellites needed	No polar coverage, Equatorial bias, Fixed position
Signal Characteristics	Minimal path loss, Lower power needs, Doppler effects	Moderate path loss, Medium power needs, Some Doppler	Highest path loss, High power needs, Minimal Doppler

infrastructure and curb hostile acts in orbit.[5] NATO's 2022 Strategic Concept elevates malicious acts that affect space assets, including cyber operations, to the level of an armed attack that could trigger collective defense.[6] Finally, a 2025 joint statement by the ITU, ICAO, and IMO urges all states to enforce radio-regulation safeguards after a surge in GNSS jamming and spoofing incidents, underscoring the need for coordinated cross-sector responses.[7]

IV. CYBERSECURITY THREAT VECTOR AND NEUTRALIZATION APPROACHES IN SATCOM NETWORKS

Modern SATCOM networks face a range of security challenges, including PL vulnerabilities, cryptographic weaknesses, and emerging threats across their three primary segments: space, ground, and link. Although satellites in orbit have limited direct physical contact, making them less vulnerable to physical sabotage, they are still susceptible to cyber threats [59] that can compromise their onboard functions and security controls [60]. The onboard subsystems of satellites are also vulnerable to hardware exploitation and control system compromises. These vulnerabilities extend into the link segment, where the communication channels between space and ground components face attacks, including signal jamming, spoofing, and eavesdropping, which exploit both physical-layer weaknesses and cryptographic limitations. Although signals from satellites in GEO, owing to their relatively large downlink beam width, can be more easily intercepted [61], the ground segment becomes the most susceptible point of entry for enemies. This is because they have accessibility and direct control over satellite operations, allowing attackers to hack GSs to control satellite operations,

[5]Reducing space threats through norms, rules and principles of responsible behaviours.

[6]NATO Strategic Concept 2022.

[7]UN agencies warn of satellite navigation jamming and spoofing.

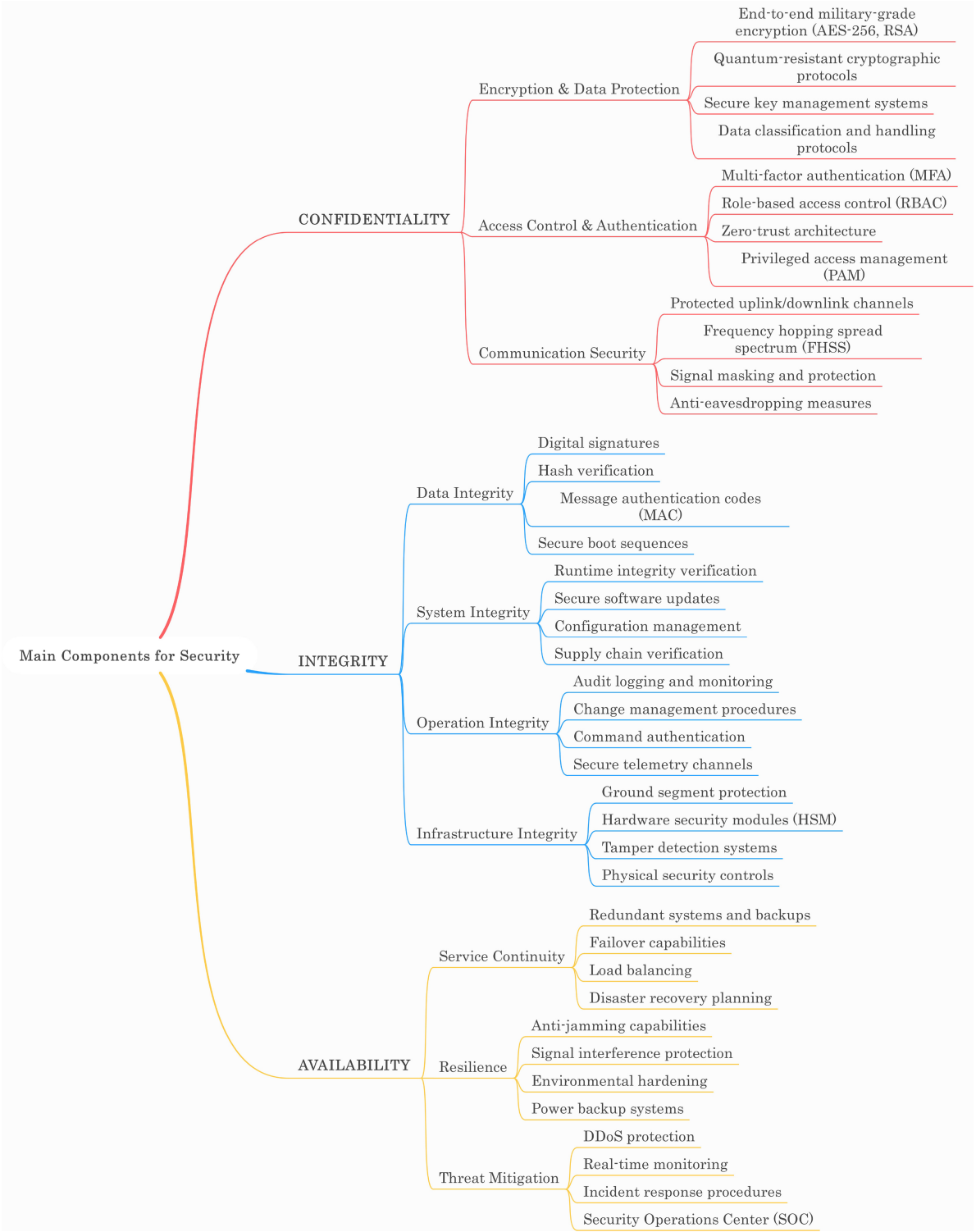


FIGURE 3. Key components of cybersecurity in SATCOMs: The CIA Framework.

insert malicious orders, or interfere with service availability by combining typical cybersecurity concerns with attack vectors that are unique to satellites [62]. The following section examines the attacks on SATCOM networks in detail.

A. PHYSICAL THREATS TO SATCOMS

These attacks exploit weaknesses in the physical transmission medium or hardware and directly disrupt SATCOM networks.

1) ENERGY ATTACKS

Energy attacks/anti-satellite (ASAT) weapons involve the disabling of satellites via physical means, mechanical interference, or invisible energy. They involve the use of high-energy lasers/forms of electromagnetic radiation to target satellite sensors, electronics, and other critical components, and are particularly effective against satellites in LEO owing to their lower altitude and smaller size. Such attacks are difficult to defend against because they are launched from a distance and do not require physical access to the satellite. Moreover, they are not limited to satellites; other spacecraft and ground-based systems can also be targeted. However, satellites are particularly vulnerable owing to their reliance on electronic systems and exposure to the electromagnetic spectrum [63].

Energy/ASAT weapons have evolved significantly since the Cold War. Early approaches ranged from the U.S.'s nuclear-tipped missile interceptors to the USSR's co-orbital ASAT weapons, which maneuvered into a target's orbit for destruction. Modern ASAT technology has evolved from crude methods to more precise and sophisticated approaches.

In the 1970s, the Soviets developed the Terra-3 [64] laser weapon, followed in the 1980s by the U.S. development of the kinetic ASM-135A ASAT missile. The latter weapon effectively eliminated the Solwind P78-1 satellite during the tests. Post-Soviet Russia developed the laser weapon Peresvet, claiming achievements in the Ukrainian conflict, and the potentially deployable Burevestnik missile [65]. Recently, several high-profile incidents involving energy attacks on satellites have occurred. For instance, in 2022, Russian military intelligence launched a kinetic attack that disrupted Viasat's KA-SAT satellite system,[8] cutting off Internet services for tens of thousands of users in Europe, including Ukraine itself. The attack used a malicious update and "AcidRain" malware to target SurfBeam 2 modems, disabling civil and military communications [11]. This incident mirrors past cyberattacks in Georgia (2008) and Ukraine (2014), which were linked to military invasions. The evolution of satellite interception systems can be summarized in terms of several key developments.

Satellite Destruction via Kinetic Impact: Target satellites are destroyed by colliding with fast-moving objects. Ground-launched ASAT missiles and space-based satellite interceptors are two examples of the kinetic ASAT weapons that have been created. Rocket propulsion is used in these weapons to drive a payload at extremely high speeds for direct contact or to create a thick cloud of metallic debris that destroys the target in orbit. China's SC-19 system, Russia's Burevestnik, and the United States' SM-3 are notable examples.

Invisible Energy Attack: These weapons employ high-energy beams, such as lasers and microwaves, to disable or eliminate targets. Directed energy weapons deliver near-instantaneous strikes, with energy traveling at or near the speed of light. The United States' MIRACL is a prominent example.

[8]A network of satellites providing internet connectivity across Europe. The American company Viasat, which serves as a defense contractor for the US government, also offers retail services and has an agreement with the Ukrainian law enforcement and armed forces.

2) JAMMING

Jamming involves broadcasting noise with sufficient power within the same frequency range utilized by the transmitter and receiver, thereby compromising the quality of the communication link [66]. SATCOMs can disrupt, hinder, or completely block by this process [13], [22]. Jamming can interfere with inter-SATCOM or even satellite-to-ground (S2G) station transmissions [60]. Various studies have emphasized the inadequacy of current communication protocols, such as code division multiple access (CDMA), when faced with potent jamming devices targeting communication channels [67]. Over the years, several jammer classes have been identified in the literature [68]. There are active and reactive jammers; the former can fall into constant, alternate, and proactive categories, given that they choose a channel in advance and jam it [69], while the latter category can adapt their jamming strategies based on the detected activity of the target communication system [70]. Moreover, a wide range of tactics can be employed for jamming. A brief classification is discussed below.

Suppression jamming: Suppression jamming employs high-power noise signals to disrupt communications by raising the noise floor and masking legitimate signals [71], [72]. The primary characteristics of suppression jamming are its broad jamming range, evident jamming impact, and ease of use and implementation. However, suppression jamming requires significant power resources owing to the noncoherent nature of the interference. Its operational effectiveness varies considerably based on communication parameters [73], signal-to-noise ratio, target distance, and environmental conditions.

Suppression jamming can be categorized into two types: broadband and narrowband jamming [74].

Broadband jamming is an interference that spans a wide range of frequencies, overwhelming the receiver's ability to filter noise. It disrupts multiple channels of communication via techniques such as barrage jamming, in which a wideband noise signal is transmitted to saturate the communication channel [75]. Pulse, matched-spectrum, and broadband Gaussian jamming are subcategories of broadband jamming.

Narrowband jamming, on the other hand, targets specific frequency bands. It is difficult to detect because it targets specific frequencies used for synchronization. Narrowband jamming can affect GNSS performance. This could lead to increased bit error rates and reduced positioning accuracy [76] which is a serious threat to 5G networks [77]. Furthermore, considering the number of frequencies jammed simultaneously, jammers can be categorized as single jammed frequency [78], sweep/multiple frequency jamming [79], and barrage frequency jamming [80].

Deception jamming: Deception jammers emit signals that generate multiple false targets that mimic real radar echoes, making it challenging to distinguish genuine signals from fake ones [81]. This technique is particularly effective for GPS receivers and synthetic aperture radar (SAR) systems [82]. In contrast to traditional jamming methods, this jamming technique achieves its objectives with lower power requirements and less detectable interference [83]. Two popular types of deception jamming are the generated and repeater types. The former produces synthetic signals that mimic legitimate navigation signals (often used in spoofing attacks), whereas the latter captures real signals, modifies them, and retransmits them to deceive the receiver.

Uplink and downlink jamming: Jamming attacks in SATCOMs can be classified as uplink or downlink [84]. The former targets payload and command signals, whereas the latter affects the downlink broadcasts and navigation signals [15]. Uplink jamming poses a significant threat owing to increased packet loss and blockage of legitimate user access across the network. This is particularly dangerous for mission-critical applications that require real-time communication [85]. Adversaries exploit the geographical proximity of jamming devices to ground stations to effectively disrupt uplink communication [70], [86]. In contrast, downlink jamming has a more localized range of influence, making it less detectable and well-suited for localized conflicts [87].

B. NEUTRALIZATION OF PHYSICAL THREATS TO SATCOMS

1) NEUTRALIZATION TO ENERGY ATTACKS/ASAT WEAPONS

Physical satellite disruption methods encompass kinetic ASAT missiles, hypervelocity object collisions, and orbital debris impacts, which create catastrophic energy-based destruction of space assets. Therefore, a multilayered defense strategy is required to prevent these attacks. Technological innovation, resilient architecture, and international policy frameworks are key defense requirements.

Prevention: Satellite Hardening and Maneuverability Satellite hardening techniques, such as radiation shielding and kinetic armor (like the ISS's Whipple shields [88]), have been essential in reducing damage from hyper-velocity collisions and electromagnetic pulses. Radiation Shielding is a standard, practical, and necessary aspect of satellite design.[9] Kinetic Armor, however, in the conventional sense (heavy physical armor), remains largely conceptual due to mass and cost constraints. Moreover, innovations such as SpaceX's Starlink network, which has autonomous thrusters to prevent collisions, demonstrate the feasibility of avoidance strategies in LEO [89]. Similarly, the Next-Gen OPIR satellites of the U.S. The Space Force uses sophisticated infrared detectors for maneuverability to detect missile launches and evade incoming threats [90]. Therefore, these state-of-the-art techniques are considered the first line of defense against physical attacks on space assets.

Defense: Detection and Interdiction Defense mechanisms designs could also serve protective purposes [22]. Two concepts, space situational awareness (SSA) and active defense systems (ADS), are paramount to this end. SSA allows the tracking of potential ASAT threats via ground-based radar systems (for example, U.S. Space Fence [91]) and space-based sensors (e.g., SBIRS [92]), thereby providing timely information for the mitigation of threats. The U.S. space fence radar system could track objects as small as 10 cm in LEO [93]. Similarly, ADS such as terminal high-altitude area defense (THAAD) interceptors and high-energy liquid laser area defense systems (HELLADS) to disable missile guidance systems during mid-flight are also central to threat-neutralization. In the past, THAAD interceptors have been successfully tested against medium-range ballistic missiles (e.g., the 2017 North Korean ICBM test).

Resilience: Redundancy and Rapid Recovery The shift towards a distributed satellite architecture has redefined resilience through redundancy. Modern satellite constellations (e.g., Starlink and OneWeb) deploy multiple satellites across orbits to create networks that can withstand individual satellite failures. These satellite mega-constellations ensure mission continuity even if individual nodes are destroyed [94]. They can leverage responsive launch systems, such as Rocket Lab's Neutron [95] to quickly launch new satellites to replace those lost, ensuring minimal downtime.

Collaboration: Policy and Deterrence Although the Outer Space Treaty (1967) [96] restricted weaponisation in space, recent initiatives, such as UN Resolution 75/36 (2020) [97] have pushed for comprehensive bans on debris-generating ASAT tests. In this wake, the ASAT testing moratorium in 2022, endorsed by the U.S., UK, and Japan, which bans destructive ASAT tests, is a positive step towards stronger SSA and debris mitigation efforts [98]. However, this endorsement came in the wake of the Nov. 2021 incident when Russia's direct-ascent (DA-ASAT) test to destroy its defunct COSMOS 1408 satellite using a Nudol ASAT missile not only produced thousands of debris pieces [99] but also endangered space assets and human spaceflight for years. Modern policy approaches have also expanded through the Combined Space Operations Initiative (CSpO), facilitating vital SSA sharing among allies. Future efforts should prioritize international collaboration to address gaps in debris mitigation and cost scalability. No single nation can unilaterally secure orbital or ground assets. Initiatives such as the UN's open-ended working group on reducing space threats [100] must evolve into binding agreements. Stakeholders must collaborate to establish standards for collision avoidance and debris mitigation. The stakes are existential because satellites underpin global communication, navigation, and security.

[9]Juno Radiation Vault.

2) NEUTRALIZATION TO JAMMING THREATS: ANTI-JAMMING TECHNIQUES

Anti-jamming research has been extended to S2G, ground-to-satellite (G2S), satellite-to-satellite (S2S), and ground-to-ground (G2G) to secure SATCOM network communications [13] against cybersecurity threats. In S2G broadband jammers target downlink signals (e.g., GPS spoofing, GNSS interference) to disrupt services for ground users (e.g., GPS, satellite TV, Internet).[10] Similarly, in G2S jamming, the uplink signals prevent the satellites from receiving commands or data.

Traditional Anti-jamming techniques: To counter this traditional approaches like spread spectrum (SS) - frequency hopping spread spectrum (FHSS); direct sequence spread spectrum (DSSS) - and their variations have been utilized since last 70 years particularly in military SATCOMs [101]. FHSS rapidly switches the transmission frequency across a wide band, making it difficult for jammers to target a single frequency. It is highly effective against narrowband jammers (NBJs) and is commonly used in military SATCOM networks. In contrast, DSSS [102] spreads signals over a wide bandwidth using a pseudo-noise code to dilute the jamming impact. This is common in GPS (e.g., GPS P(Y)code) and is effective against wideband jammers (WBJs). Apart from SS, techniques such as beam forming with phased-array antennas, null steering, and multi-constellation receivers have been used to mitigate narrowband and WBJs. However, these methods rely heavily on hardware to avoid or overpower jamming signals.

Jamming statistics-dependent frequency hopping (JSDFH) [103] represents a paradigm shift with a proactive approach to jamming. This adds a layer of intelligence to traditional FHSS. The RF environment was monitored to collect data on jamming patterns, frequencies, and timing. This data was used to build a statistical model to predict the likely jamming behavior. The system then designs frequency-hopping sequences that avoid high-risk frequencies and adapt to changing jammer tactics based on a statistical analysis [104].

Advanced anti-jamming techniques: Although FHSS and DSSS are foundational anti-jamming techniques, their effectiveness is limited by their reliance on pre-shared keys, vulnerability to reactive and intelligent jamming, and implementation complexity [105]. Modern research focuses on overcoming these limitations using techniques such as dynamic, AI/ML-driven, and adaptive anti-jamming. They leverage software-based solutions such as real-time signal processing algorithms to dynamically adapt to jamming threats. ML techniques can be applied to a variety of SATCOM issues [46], [106], including interference removal, optimization of radio resources (power and spectrum), optimization of SATCOM network operation, and management of large satellite constellations [34]. Research [106] is part of the application portfolio of a newly developed ML framework for space missions.

Cognitive radio (CR) has emerged as a modern AI-driven approach to spectrum management, including the detection and mitigation of jamming. It uses convolutional neural networks (CNN) [107] or reinforcement learning (RL) [108], [109]. Techniques such as energy detection [110], [111], cyclostationary feature detection [112], [113], and matched filtering [114] are used for spectrum sensing and jammer identification. Major initiatives in CR have shown significant advancements in anti-jamming. For instance, the Defense Advanced Research Projects Agency (DARPA) has spectrum-focused programs. These programs include RF machine learning systems (RFMLS)[11] which develops ML algorithms for real-time RF analysis and signal classification. Similarly, communications in contested environments (C2E) have developed technologies that ensure reliable military SATCOMs in environments with heavy jamming and electromagnetic interferences. These programs collectively advance CR capabilities through autonomous spectrum management, multiband operation across the L-band, C-band, and military frequencies, and adaptive interference mitigation.

Fig. 4 provides brief and non-exhaustive classical and modern anti-jamming techniques in SATCOMs. This classification highlights a clear shift from traditional static anti-jamming techniques to adaptive, AI-driven solutions. Although FHSS, DSSS, and beamforming remain fundamental satellite anti-jamming techniques, recent advancements emphasize CR, RL, and game theory-based optimizations [115], [116] to mitigate jamming threats in real time. However, advanced AI/ML solutions require computational overhead and are energy intensive.

C. CONFIDENTIALITY AND INTEGRITY THREATS TO SATCOMS

1) EAVESDROPPING

Eavesdropping refers to the covert acquisition and surveillance of private conversations, electronic messages, or transmissions between individuals without their knowledge or consent [22]. The broadcast nature of the wireless medium via RF, extensive coverage areas, unencrypted communications resulting from low bandwidths [13], [44] and a large number of LEO satellites [117] have exacerbated the situation, leading to increased eavesdropping attacks from both terrestrial and aerial platforms and resulting in breaches of sensitive information [118]. However, the consequences of eavesdropping are far more serious and extend beyond individual privacy concerns, impacting national security and public safety [122], [123]. Eavesdropping attacks are categorized as passive or active. Table 5 provides a brief overview of both. Passive eavesdropping mostly

[10] Black Sea GPS Spoofing Incident (2017). Iranian GPS jamming in the Persian Gulf (2019–2021). Chinese Anti-satellite (ASAT) Test (2007). North Korean GPS jamming (2016). Jamming of Starlink in Ukraine (2022–2023).

[11] https://www.darpa.mil/research/programs/radio-frequency-machine-learning-systems

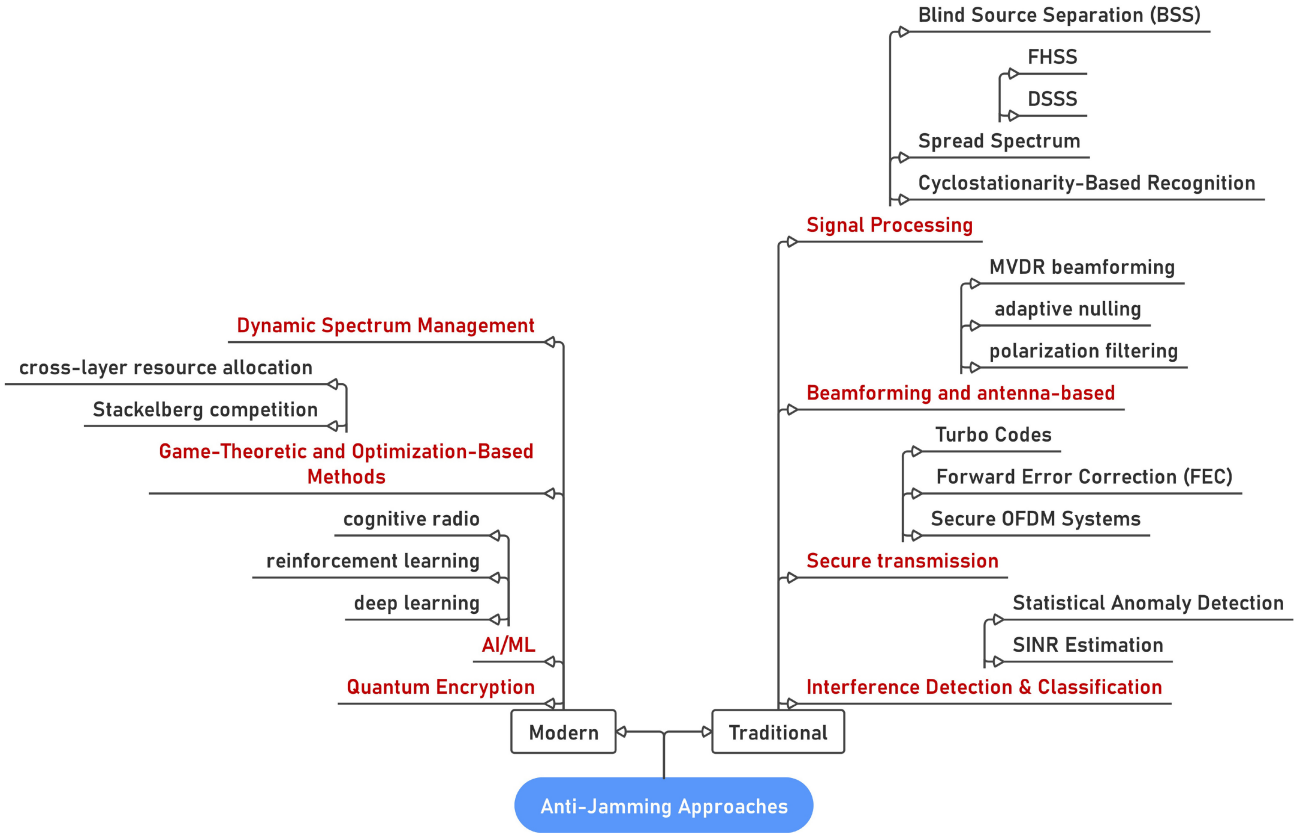


FIGURE 4. A brief classification of anti-jamming techniques in SATCOMs.

TABLE 5. Comparison of passive and active eavesdropping in SATCOMs.

Characteristic	Passive Eavesdropping	Active Eavesdropping
Definition	Listening to or recording communications without altering transmission.	Directly intercepting and potentially modifying the communication.
Victim Awareness	It is impossible for the victim to detect.	Possibility of detection by the victim.
Impact on	Personal information.	Cryptography (decryption).
Technical Difficulty	Easy.	Requires sophisticated hardware or software.
Mitigation	Easy.	Hard.
Vulnerability Exploitation	Exploits weaknesses in satellite downlink signals [119].	Can leverage open-source software vulnerabilities in GMR-1 and GMR-2 SCS standards [120].
Information Obtained	Can identify individual satellite customers and their activities [121].	Enables decrypting live call sessions using satellite networks [120].
Operational Challenges	Requires access to satellite communication channels.	Can be achieved with common, moderately priced equipment [120].

involves listening to transmissions without altering them. It is undetectable, technically simple, and targets personal information theft. Passive eavesdropping exploits downlink vulnerabilities and requires access to the communication channels. Active eavesdropping, on the other hand, modifies communication and is detectable. It targets encrypted content and requires advanced tools to detect. Attackers can exploit known software flaws to intercept and decrypt the live sessions.

2) SPOOFING

Spoofing in SATCOMs refers to an attack in which a malicious actor transmits false signals/data to deceive a

TABLE 6. Comparison of spoofing types in SATCOMs.

Type of Spoofing	Target	Method	Tools/Techniques	Exploited Vulnerabilities
GNSS/GPS Spoofing	Navigation systems (e.g., drones, ships)	Broadcast counterfeit GPS/GNSS signals to manipulate location or time data, including: - **Signal Meaconing and Replay/-Forwarding Spoofing Attacks:** Delay and retransmit genuine signals, manipulating code phase or timing. - **Signal Estimation and Replay/-generating spoofing attack:** Estimate missing parts of signals (e.g., cryptographic codes) and replay with a delay or offset. - **Signal Modification:** Nulling signals are used to override authentic signals, inducing false positions or timing fixes.	- Software-defined radios (SDRs) - High-gain antennas	- Weak civilian codes (e.g., GPS C/A code) - Code phase and timing vulnerabilities - Weak cryptographic protections - Inability to detect nulling signals.
TT&C Spoofing	Satellite control systems	The forging of command signals to hijack satellite operations.	- Protocol reverse-engineering - Signal generators	- Unencrypted command channels - Weak session keys
IP Spoofing over Satellite Links	Satellite internet services (e.g., VSAT)	Mask source IP addresses to bypass authentication.	- Packet crafting tools (e.g., Scapy) - Botnets	- Lack of source validation - Weak firewall rules
Replay Attacks	Authorization systems (e.g., satellite TV)	They can record and retransmit legitimate signals to bypass security checks.	- Signal recorders - Delay-and-replay devices	- Unencrypted channels - Static session tokens
Hybrid Spoofing	Multilayered systems (e.g., military comms)	Combining jamming (DoS) with spoofing to overwhelm the receivers.	- Jammers + SDRs - Directional antennas	- Lack of signal diversity - Poor anomaly detection

receiver by impersonating a legitimate satellite, GSs, or user. This misleads one or more intended recipients through the use of delayed, fabricated, or inaccurate signals [124]. Attackers create fake signals that match the frequency, modulation, and protocol of authentic systems (e.g., GPS L1 at 1575.42 MHz) [125]. High-gain antennas/amplifiers are used to overpower authentic transmissions, and these spoof signals are synchronized with real signals in terms of timing, Doppler shift, and cryptographic parameters (if applicable).

Spoofing attacks can have severe consequences. They can cause safety issues, such as navigational problems in air and marine operations, as demonstrated by the 2017 Black Marine incident, in which over 20 vessels reported incorrect GPS positions [126]. Similarly, they can endanger national security, as evidenced by Iran's 2011 seizure of an American military drone [126]. Spoofing disrupts sectors that rely on exact placement, causing delays and financial losses. Table 6 presents a brief comparison of various spoofing techniques in SATCOMs. Five distinct spoofing types that affect SATCOMs were identified in this study. GNSS/GPS spoofing attacks target navigation systems by broadcasting counterfeit signals. Attackers employ SDRs with high-gain antennas to delay or modify the signals. These attacks exploit the weaknesses of GPS C/A codes and the timing systems. TT&C spoofing compromises satellite control using fake commands. Attackers use signal generators and reverse-engineered protocols to exploit unencrypted channels. IP spoofing conceals the sender's identity on VSAT links. Attackers use botnets and packet manipulation tools to bypass authentication using fake IP packets. The lack of source validation creates security vulnerabilities. Replay attacks capture legitimate transmissions for reuse. This method bypasses security by exploiting unencrypted channels and static authentication tokens. Hybrid spoofing combines jamming and spoofing techniques to simultaneously disrupt and mislead systems. To this end, directional antennas help attackers exploit the poor signal diversity. Each attack type targets a specific system and exploits specific technical vulnerabilities.

D. NEUTRALIZATION TO CONFIDENTIALITY AND INTEGRITY THREATS OF SATCOMS

1) ANTI-EAVESDROPPING

Anti-eavesdropping methods at physical layer: Typical anti-eavesdropping strategies at PL consist of beamforming, adding artificial noise (AN), cooperative jamming, and

other strategies [127]. Conventional beamforming remains a central component of PLS in SATCOMs. In antenna arrays, beamforming is employed to control the signal directionality and focus transmissions in specific directions. This makes eavesdropping significantly more challenging [22]. Various beamforming strategies have been explored, from cooperative schemes protecting Ka-band fixed satellite service (FSS) terminals [128] to sophisticated multibeam systems that actively nullify eavesdropper signals [129] and partial zero-forcing techniques that orthogonalize legitimate user signals against multiple eavesdroppers [130]. Dynamic rotated angular beamforming (DRAB) further enhances PLS in mmWave communications [131]. Unlike conventional methods, it dynamically rotates the main lobe and suppresses side lobes. This provides a robust, equipment-free security solution that adapts to real-time eavesdropping threats.

Two additional techniques bolster PLS against unauthorized interception: adding an AN and using ambient backscatter communication (ABSC). ABSC reuses existing signals/reflections to covertly transmit information, which can also be leveraged to confuse eavesdroppers. AN increases the channel capacity for legitimate users while degrading it for eavesdroppers [132]. This involves generating noise that coincides with the signal space of potential eavesdroppers, making it difficult for them to decode the communication. To this end, DNN-based precoding optimization for MISO wiretap channels has been employed via AI to optimize AN placement and enhance security across the entire channel [133]. Similarly, Ding et al. [134] utilized relay-assisted AN transmission for two-way relay networks, actively transmitting an AN during communication to counter side-lobe eavesdropping and enhance the security. ABSC [135] involves strategically transmitting a part of the message through conventional active transmission. This exercise attracts and drains the eavesdropper's energy while simultaneously backscattering the remaining message using the transmitter's own signals.

Note that DNN-based precoding optimization also faces critical limitations. This includes performance degradation under imperfect channel state information (CSI) conditions because many schemes assume an unrealistic perfect channel estimation [133]. The computational complexity of training DNNs for large-scale MISO systems presents significant challenges, particularly with nonlinear precoding optimization [136].

ML/DL-based methods: ML and deep learning (DL) methods to counter intrusion/eavesdropping in SATCOMs are primarily categorized as detection and prevention based methods. The former category analyzes a signal for anomalies or relies on feature selection to enhance security [137]. Prevention-based methods largely focus on encryption and optimal access strategies to maximize secrecy rates in space-air-ground networks (SAGN) [138]. Detection methods include hybrid intrusion detection [139], deep hierarchical networks [140], and GRU-based systems for detecting cyber-physical and network security threats [141]. The objective is to identify vulnerabilities across satellites [142], unmanned aerial vehicles (UAV), and cellular networks using advanced anomaly detection ML/DL algorithms [143]. Federated learning (FL) frameworks have also made strides in detecting intrusions in satellite-terrestrial integrated networks (SAT-IDSs). For instance, Jiang et al. [144] combined conditional generative adversarial networks (CGANs) and BiLSTM for intrusion detection in SAT-IDSs [146]. Similarly, the robustness of a system is assessed by simulating adversarial attacks using lightweight gradient evaluation [147]. Distributed intrusion detection mechanisms for satellite-IoT frameworks using an SDN backbone have also been used [148].

All studies [138], [139], [140], [141], [142], [143], [144], [146], [147] presented are primarily conceptual, with experimental implementations limited to simulations or controlled laboratory environments using existing datasets (such as CIC-IDS2017, UNSW-NB15, NSL-KDD).

limitations of ML/DL-based methods: ML and DL solutions for SATCOM security still face three critical gaps. *(i) Dataset scarcity and external validity:* most intrusion-detection studies are trained on laboratory traces or synthetic waveforms, so models that reach 95–99% accuracy in simulation often collapse when confronted with live, multi-orbit traffic or atypical modulation schemes [139], [140], [142], [144]. *(ii) Limited generalizability:* a classifier tuned for one satellite, band, or constellation seldom transfers to another because the underlying feature distributions differ; the problem mirrors the non-IID data and unseen-class hurdles documented in federated-learning research, where the AZSL framework had to generate synthetic samples just to keep a global model robust across heterogeneous clients [145]. Generalizability is indispensable in SATCOM because a security stack must protect GEO relays, LEO broadband satellites, and diverse ground stations without per-platform retraining, which is infeasible over narrow telemetry links. *(iii) Resource and threat constraints:* A satellite cannot host large CNN or transformer models, and their lightweight surrogates remain vulnerable to adversarial perturbations that flip classifications with imperceptible noise; moreover, distributed learning methods such as federated or split learning introduce synchronization overhead and new attack surfaces when applied across heterogeneous SATCOM segments. Closing these gaps will require shared real-world benchmarks, domain adaptation or zero-shot techniques that learn transferable representations, and ultra-compact yet attack-resilient architectures tailored to radiation-hardened hardware.

Cryptographic anti-eavesdropping methods: SB-QKD secures the physical layer by encoding random bit strings onto the quantum states of single photons or qubits and transmitting them between a satellite and optical ground stations [149], [150]. Protocols such as Bennett–Brassard-84 (BB84) and SARG04 leverage two fundamental quantum properties: the no-cloning theorem, which forbids the perfect duplication of an unknown qubit state [151], and measurement disturbance, whereby any observation of a qubit

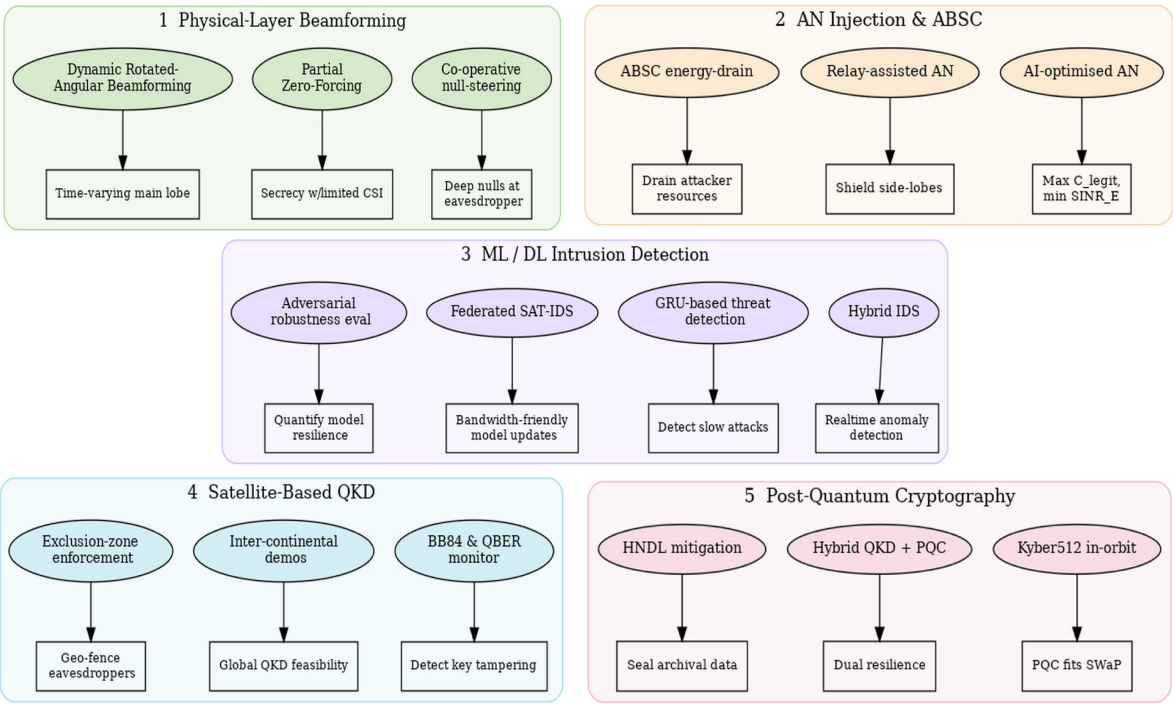


FIGURE 5. A brief summary of key takeaways in anti-eavesdropping techniques.

irreversibly alters its state. After the quantum exchange, the satellite and ground station use a public but authenticated classical channel to sift matching measurement bases, estimate the quantum bit error rate (QBER), and apply error correction and privacy-amplification routines. If an eavesdropper taps the free-space optical link, the induced perturbations increase the QBER beyond an agreed threshold, alerting the legitimate nodes to discard the compromised key material [149], [153].

In the SATCOM context, SB-QKD overcomes the ∼100 km loss limit of terrestrial fiber by exploiting the low-attenuation line-of-sight paths achievable from low-Earth-orbit (LEO) or geostationary (GEO) platforms. Thus, a satellite can distribute fresh symmetric keys to widely separated ground stations, create exclusion zones that localize potential eavesdroppers through QBER monitoring, and refresh higher-layer cryptographic suites used for telecommand, telemetry, or end-user broadband links. Demonstrations such as *Micius* (China) and the forthcoming *QEYSSat* mission (Canada/Europe) have already shown intercontinental key exchange with secret-key rates sufficient to seed one-time pad or AES session keys, validating SB-QKD as a practical pillar for future satellite network security [154].

The use of SB-QKD in conjunction with NIST-initiated post-quantum cryptography (PQC) can further fortify SATCOMs. Quantum algorithms such as Shor's [155] and Grover's [156], which are capable of decrypting data in polynomial time [157] have exposed the weaknesses of traditional cryptographic algorithms. Therefore, the integration of PQC is essential to safeguard against future quantum decryption attacks (for example, the Harvest Now, Decrypt Later (HNDL) [158]). The PQC protocol Kyber512 was successfully integrated into the KEMTLS-PDK for secure session establishment between GSs and ESA's OPS-SAT, demonstrating feasibility in orbit [159]. Fig. 5 highlights the key takeaways in anti-eavesdropping techniques.

2) ANTI-SPOOFING MEASURES

A spoofing attack requires precise control over waveform replication, power manipulation, signal synchronization, and Doppler shift compensation to deceive the receiver [160]. Attackers leverage SDRs to generate false signals that mimic authentic GNSS signals in terms of their phase, power, and timing. Over the years, numerous anti-spoofing measures have been proposed to combat these attacks. Some methods rely on analyzing the signal parameters, whereas others use specialized hardware. Contemporary research has focused on ML/DL methods and cryptographic techniques. A brief overview of these anti-spoofing strategies is provided in the following sections. Whereas, Fig. 6 highlights the key takeaways in anti-spoofing techniques.

Signal characteristic analysis: Signal characteristic analysis methods identify anomalies in the signal direction, range variations, and frequency shifts.

Angle-of-arrival (AOA) estimation is a key source for detecting and mitigating spoofing attacks. Spoofed signals from a single terrestrial source have a consistent AOA across multiple satellites, whereas authentic satellite signals arrive from different angles due to orbital motion This

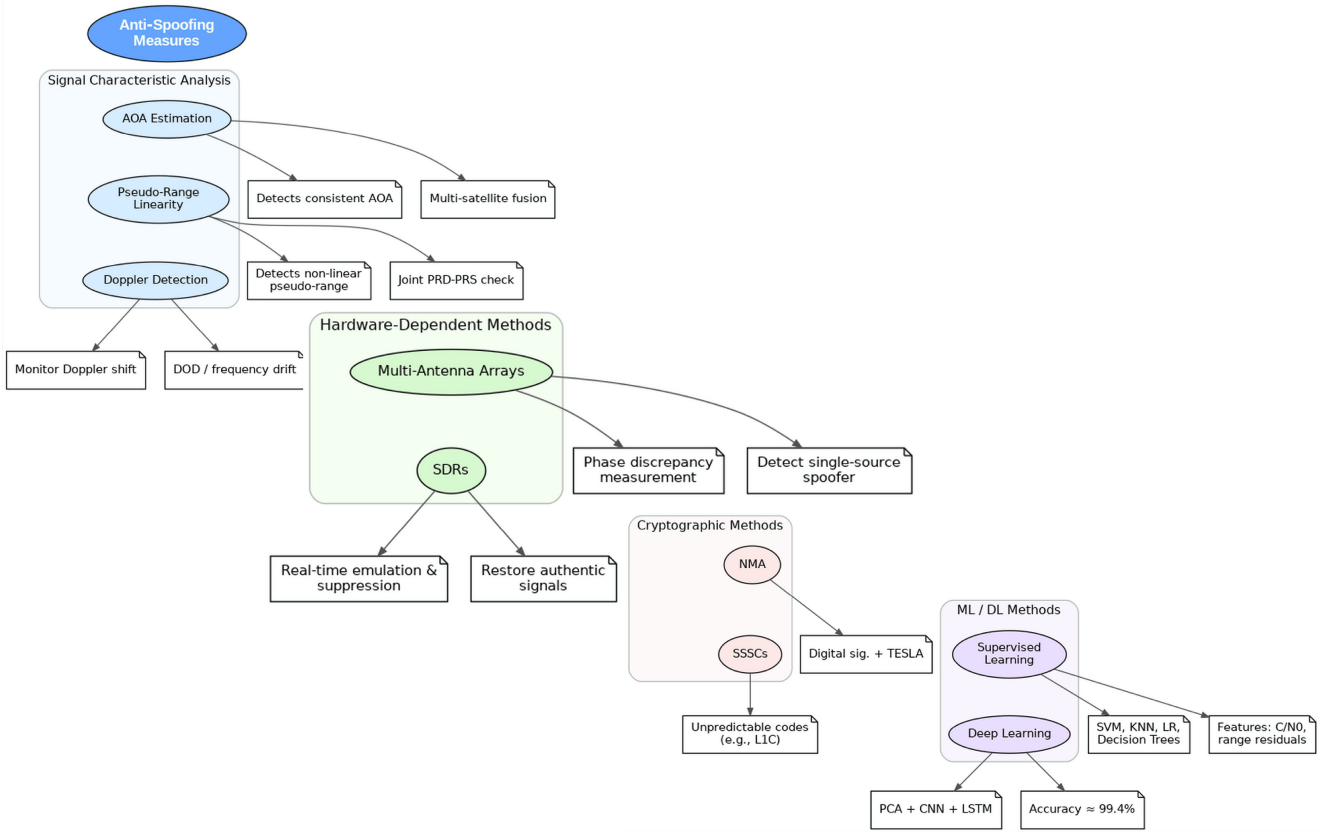


FIGURE 6. A brief summary of key takeaways in anti-spoofing techniques.

inherent difference forms the foundation of AOA-based spoofing detection methods. For instance, Psiaki et al. [161] measured carrier-phase differences using dual antennas; authentic multi-satellite signals versus a single-source spoof yielded inconsistent phase patterns, revealing the spoofing. Similarly, direction-of-arrival (DOA) estimation categorizes signal sources by elevation angle [162], helping to distinguish a single-direction spoofing transmitter from legitimate signals that come from multiple directions. Multi-satellite fusion techniques have been utilized to further improve accuracy. Such techniques weigh the measurements from each satellite (azimuth and elevation) by the carrier-to-noise ratio C/N_0 uncertainty; signals with a more certain C/N_0 contribute more [163]. This refinement reduces the number of false positives in detection.

Pseudo-range linearity monitoring is another approach against spoofing attacks. When a receiver moves (e.g., in a vehicle or aircraft), the spoofing signals transmitted from a single ground-based source exhibit nonlinear pseudorange variations. This nonlinearity arises because the spoofing source cannot accurately simulate the geometric relationships and relativistic Doppler shifts of multiple satellites in motion across different orbits. In contrast, authentic signals from genuine satellites maintain linear pseudo-range differences (PRD) and pseudo-range sums (PRS) owing to the predictable motion of the satellites and the receiver's trajectory. The joint PRD-PRS linearity monitoring method was validated in both uniform and variable motion scenarios to detect spoofing [164].

The Doppler effect occurs when the wave frequency changes according to the relative motion between the source and observer. Satellites typically travel at very high speeds relative to the receiver, resulting in a Doppler shift. A matching Doppler shift between the genuine satellite signal and the receiver's signal indicates normal operation; a significant discrepancy may indicate a spoofing attack Multiple studies have demonstrated Doppler-based spoofing detection, proposing methods such as Doppler offset detection (DOD) [165]] and monitoring Doppler frequency drift [166].

Hardware-dependent methods: These anti-spoofing measures rely on specialized hardware such as multi-antenna arrays (MAA) to detect or suppress spoofing. The role of the MAA in resisting spoofing attacks on GNSS receivers is crucial. Two or more antenna systems measure phase differences to identify potential spoofing sources, and discrepancies in the phase between antennas serve as indicators of spoofing activity [167], [168].

In addition, SDRs enable real-time spoofing emulation and suppression capabilities. Stochastic gradient descent (SGD) algorithms project the received signals onto spoofing-orthogonal subspaces to restore authentic GPS signals [169].

TABLE 7. Trade-offs among machine-learning spoofing-detection approaches in SATCOM/GNSS contexts.

Approach	Algorithms / studies	Accuracy	Compute & data demand	Deployment status
Supervised/classical ML	SVM [174], cost-SVM [175], logistic regression, k-NN, decision tree [176]	90–97%	Low; embedded GNSS chips	Field-validated only in [175]
Supervised/deep learning	PCA + CNN + LSTM [177]; MLP & dual-CNN [178]	Up to 99.4%	High; GPU or cloud off-load	Field tests in [177]

Cryptographic methods: The two most popular anti-spoofing cryptographic measures are navigation message authentication (NMA) and spread-spectrum security codes (SSSCs). In 2003, Scott introduced a pioneering anti-spoofing measure that utilized secure satellite communication codes (SSSCs) [124]. The SSSC uses unpredictable codes within GPS signals, such as L1C, which require cryptographic keys for validation. Moreover, these codes operate at high chipping rates of 1.023 MChips/s, making it exceedingly difficult for attackers to replicate them in real-time. In contrast, NMA ensures that the navigation data received by a GNSS receiver are genuine and have not been corrupted. One instance of NMA is Galileo's Open Service NMA (OSNMA), which includes cryptographic safeguards via the TESLA-based delayed-key disclosure method [170]. The OSNMA validates signal authenticity using digital signatures and shorter message authentication codes (MACs), thereby limiting spoofing by certifying unaltered transmissions from legitimate satellites. Annually updated public keys are transmitted via over-the-air re-keying (OTAR) and certified by the European GNSS Service Center (GSC), assuring cryptographic agility while minimizing bandwidth overhead.

Recent advances include hybrid authentication chains that combine elliptic curve digital signature algorithm (ECDSA) signatures for root key validation with TESLA chains [171]. This approach balances efficiency, backward compatibility, and lightweight cryptography. ECDSA is suitable for satellite links because its compact keys and signatures conserve bandwidth and reduce latency [172]. These signatures secure command-and-control messages, stopping unauthorized uplinks that could jeopardize the spacecraft. Timed Efficient Stream Loss-tolerant Authentication (TESLA) addresses broadcast traffic by revealing keys only after messages are received; thus, low-power user terminals can authenticate streams without verifying a digital signature on every packet [173]. It forms a hash-linked key chain released on a fixed schedule, which has already been proposed to shield navigation signals from spoofing and replay attacks. Together, concise ECDSA roots and bandwidth-frugal TESLA chains provide SATCOM networks with strong yet practical authentication under tight space- and ground-segment constraints.

ML/DL methods: Several ML and DL techniques have been proposed for spoofing attack detection, including the implementation of classical supervised learning algorithms such as support vector machines (SVMs). Zhu et al. [174] considered three different spoofing scenarios and utilized a combination of features, such as the carrier-to-noise ratio C/N_0 and pseudo-range residuals, to train the SVM model. Similarly, cost-SVM, a variation of the original SVM model, trained on a hybrid dataset and cross-validated against real-world datasets, demonstrated enhanced detection accuracy against GNSS spoofing/meaconing attacks[175]. Nabi et al. [176] considered a suite of algorithms: logistic regression, SVM, K-nearest neighbors (KNN), and decision trees. The models were trained using signal-tracking data against sophisticated GPS spoofing attacks, specifically the secure code estimation and replay (SCER) type.

In the DL domain, an amalgamation of principal component analysis (PCA), CNN, and LSTM has been proposed for dimensionality reduction, spatial feature extraction, and temporal dependency modeling, respectively [177]. The system achieved an overall accuracy of 99.43%. Similarly, Borhani-Darian et al. [178] explored the cross ambiguity function (CAF) to distinguish between real and spoofed signals. A multilayer perceptron (MLP) and two types of CNNs were trained using simulated data. The results demonstrate that these DL models can effectively identify spoofing attacks. Among the reviewed papers, [174], [176], [178] are predominantly conceptual, relying on simulated or controlled datasets without a real-world deployment. In contrast, [175], [177] incorporated real-time or field-acquired datasets. Table 7 provides a brief contrast of different neutralization methods (Classical ML vs. DL methods for spoofing detection). This will help readers understand the trade-offs between accuracy, complexity, and deployment feasibility.

E. AVAILABILITY THREATS TO SATCOMS

1) DOS AND DDOS

A DoS attack overwhelms a satellite system (e.g., GSs, satellite modems, or uplink/downlink channels) to disrupt its availability. A distributed denial-of-service (DDoS) is a coordinated DoS attack launched from multiple distributed sources (e.g., botnets) to amplify the impact and evade mitigation. DoS/DDoS attacks exploit the unique vulnerabilities of satellite systems. For instance, RF jamming disrupts communication by overwhelming satellite frequencies, such as the L-band, using tools such as software-defined radios (SDRs) and jamming pods. DoS/DDoS attacks on satellite

TABLE 8. DoS/DDoS mitigation techniques in SATCOMs.

Key Principle	Implied Technique	Description	Attack Vector Analysis	References
Adaptive traffic monitoring	ICMP/UDP flood mitigation; real-time analysis	Block or limit malicious traffic and prioritize legitimate flows using dynamic thresholds.	DoS/DDoS	[179], [180]
SDN architectures	Centralised traffic control; ML integration	Dynamically reroute or block malicious flows and deploy lightweight CNNs for pattern analysis.	DoS/DDoS detection & mitigation	[181], [182]
Blockchain-based defenses	Tamper-proof data integrity; decentralized PKI; smart-contract intrusion detection	Blockchain is used to secure data exchange, enforce decentralized authentication, and detect intrusions via token-based contracts.	DoS/DDoS mitigation & detection	[40], [183] [184]
Moving target defense (MTD)	Dynamic network reconfiguration; attack surface minimization	Randomise IP/MAC addresses of critical nodes to disrupt adversary reconnaissance and mitigate attacks.	DoS/DDoS detection & mitigation	[185]

systems can disrupt critical services and cause significant losses. In 2019, Iranian-backed groups targeted Arabsat frequencies and jammed Kuwaiti satellite TV broadcasts. Similarly, in the 2022 Viasat KA-SAT attack, Russian actors used the AcidRain malware alongside DDoS techniques to cripple Ukraine's satellite Internet service.

F. NEUTRALIZATION TO AVAILABILITY THREATS OF SATCOMS

A single DoS event can ripple through an entire constellation and its ground infrastructure, as has been witnessed by Feb. 2022 KA-SAT incident.[12] Such incidents warrant multilayered DoS/DDoS mitigation approaches tailored to constraints such as latency, bandwidth, and protocol vulnerabilities. Contemporary literature on DoS/DDoS mitigation strategies suggests robust authentication and authorization frameworks that block rogue terminals before injecting traffic. Asif et al.'s [186] two-phase handshake reduces latency by 45% while remaining lightweight enough for satellite modems, and similar multi-factor schemes tailored to distributed ground stations have been demonstrated by Ansong et al. (2024) and others [187]. At the teleport and network operation tiers, adaptive rate-limiting mechanisms, such as Usman et al.'s [179] ALBDM cap on ICMP traffic, maintain near-zero packet loss during simulated floods; however, operators must tune thresholds carefully to avoid false positives during legitimate surges. The primary deployment challenges include managing authentication across distributed ground stations, maintaining security without excessive latency, and ensuring interoperability across heterogeneous satellite systems with varying security implementations. ML intrusion detectors add a third line of defense; models trained on synthetic satellite traffic now exceed 96% accuracy, even under meteorological noise, and can quarantine malicious flows before narrow feeder links are saturated [188]. If attack volumes still overwhelm ground stations, cloud scrubbing centers absorb and cleanse traffic at scale, although the detour imposes bandwidth and latency costs that must be weighed against the protection gained from the detour. Recent research has also proposed SDN-based detection. SDN approaches enable rapid path rerouting and segment isolation, allowing operators to quarantine compromised links without interrupting service, an approach that has already been validated in prototype LEO networks despite the integration challenges of proprietary architectures [189].

Table 8 provides a brief review of the methods aimed at neutralizing DoS/DDoS attacks. The table highlights the key DoS/DDoS mitigation strategies used in SATCOMs. Each method follows a specific architectural approach. For instance, adaptive traffic monitoring blocks harmful traffic and prioritizes legitimate flows using real-time analysis and ICMP/UDP flood control. SDN-based methods use centralized traffic control and ML to reroute the attacks and detect abnormalities. Blockchain solutions ensure secure data exchange, prevent spoofing, and support intrusion detection using token-based smart contracts [190] and PKI. Finally, the MTD changes the IP and MAC addresses of the network nodes to disrupt attacker tracking and delay attack planning. Each strategy aims to reduce exposure and ensure service continuity.

V. FUTURE RESEARCH DIRECTIONS

Cyber threat neutralization in SATCOMs has evolved across interdisciplinary domains, blending innovation and urgency. At the PL, the objective is to exploit RF signals, optical noise, or even quantum fluctuations to create safe wireless environments. These measures are augmented by AI/ML and cryptographic techniques to create a multilayered defense mechanism. AI/ML enhances threat intelligence and enables

[12] A Mysterious Satellite Hack Has Victims Far Beyond Ukraine.

real-time anomaly detection, whereas cryptographic advancements, such as lattice-based and hash-based algorithms, fortify resilience against emerging quantum decryption threats. This convergence of physical-layer security measures, AI techniques, and quantum-resistant cryptography creates a dynamic adaptive defense. This enables SATCOM networks to withstand sophisticated attacks. In this section, we briefly set priorities for research and development that are essential for stakeholders seeking to improve the security and functionality of SATCOMs. A number of study fields should be prioritized, given their maturity and rate of technological progress.

- Protocol Standardization: Although there are various standard protocols–CCSDS, 3GPP NTN, GSE–they are not usually used exclusively. Many systems use proprietary protocols. This mix creates a problem: they do not always communicate well with each other. Therefore, the development of integrated protocols is crucial. This enhances overall communication efficiency and interoperability by enabling various space communication systems to "talk" to one another efficiently.
- Policy and regulatory frameworks: There is a need for international collaboration in securing SATCOMs in three key areas. First, globally harmonized spectrum allocation policies should be developed to prevent interference and prioritize security-critical applications. Second, quantum-resistant mandates, including timetables for migrating to PQC, should be implemented to protect against future quantum computing risks. Finally, international frameworks for encrypted SATCOM data sharing that comply with the GDPR (EU), CCPA (US), and China's data security laws should be created.
- Public-private partnership (PPRs): PPRs have become increasingly important in the fast changing field of cyber-security as means of cooperative interventions tackling complex technical risks. For instance, a joint venture between Honeywell and the ESA, the quantum key distribution satellite (QKDSat) project, shows how intersectoral cooperation can produce state-of-the-art encryption technologies that reduce risks in data exchange and strengthen the resilience of essential infrastructure. The involvement of various stakeholders also allows real-time exchange of jamming/spoofing signatures.

VI. CONCLUSION

Space assets and SATCOMs enable a wide range of scientific, military, and commercial applications. Space has become the technological foundation for numerous sectors, including medicine, education, research, administration, and transportation. Therefore, ensuring the continuity and resilience of its services is of utmost importance for the proper functioning of a digital-networked society. The disruption of these services can cause significant financial and material losses, including the collapse of multiple digital services. Traditionally, space assets have been at risk from natural phenomena such as space debris, weather, and radiation. However, the past several decades have witnessed an exponential rise in human-made disruptions, including cyberattacks, nuclear, and ASAT weapon tests. Such vulnerabilities have been expanding across the space, ground, and link segments of SATCOM networks. In these circumstances, it is necessary to adopt adequate protection measures and resilient frameworks at both technical and legal levels.

This study analyzes both traditional and modern cybersecurity threats to SATCOM networks and proposes effective neutralization measures. It proposes a multifaceted security blueprint across the regulatory, operational, and technological domains to address cyber threats. Furthermore, focusing on lightweight AI, quantum-resilient encryption, and cross-layer security architectures is central to threat-neutralization. SATCOM operators can deploy lightweight AI models on satellites. These models detect spoofing and jamming in real time using simple classifiers and small neural networks. More complex analyses can be performed at ground stations using DL and multisource data. Other measures can also be taken to help the industry adapt to the research findings. These include the standardization of cybersecurity frameworks, such as the FCC 2024 regulations, dedicated funding for legacy system modernization, and the prioritization of quantum-safe upgrades and collaborative threat-sharing networks. International cooperation should also be expanded through treaties, such as the Artemis Accords, with binding cybersecurity provisions. A multifaceted approach promises enhanced threat detection, protocol agility, and resistance to zero-day exploits in increasingly complex satellite-terrestrial networks. This study lays a scalable foundation for the real-time adoption of public and private stakeholders tasked with securing next-generation SATCOM infrastructure.

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ADITI RANJIT KUMAR VERMA (Senior Member, IEEE) was born in Mumbai, India. She received the B.S. degree in electronics engineering from Mumbai University, Mumbai, in 2011 and the M.S. degree in electronics engineering from The University of Texas at Arlington, Arlington, TX, USA, in 2013.

Her major fields of study include cellular telecommunications, next-generation wireless networks, and network architecture design. She joined the T-Mobile, Bellevue, WA, USA, in 2016, where she is currently a Senior Systems Design Engineer. She has led major projects, including the implementation of Rich Communication Services (RCS) over 5G, containerization of messaging platforms (SMS, MMS, and RCS), and development of emergency 911 messaging using satellite communication in partnership with SpaceX. She has also contributed to centralized voicemail database solutions, regulatory compliance for messaging archival services, and telecom data monetization strategies. Her contributions have been recognized through multiple Peak Award nominations and a T-Mobile IMPACT Award from 2024 to 2025 nomination for driving business impact and innovation.